

Using SPSS to calculate Descriptive statistics (frequencies, measures of central tendency, and measures of variability)

STC224

Learning Objectives

- Frequencies Procedure
- Measures of Central Tendency and Variability
- Analysis of groups using Means Procedure
- Means procedure using the layered option

Introduction

- **Frequencies** refer to the number of observations in each category of a variable.
- If, for example, for a variable gender there were four males and six females in the data set, the frequency reported in SPSS for males would be 4 and for females would be 6.
- Frequencies are typically obtained on **categorical variables**.

Introduction cont'd

- Measures of central tendency are used to describe the center (or central location) of a set of scores and consist of the **mean**, **median**, and the **mode**.
- The **mean** is the arithmetic average (the sum of the scores divided by the total number of scores),
- the **median** is the middle score (assuming the scores were ordered from lowest to highest), and
- the **mode** is the most frequently occurring score (or scores) in the data set.

Introduction cont'd

- Measures of variability are used to describe the amount of spread or variability in a set of scores.
- The **standard deviation** and **variance** are two of the most commonly used measures of variability.
- The **standard deviation** is a measure of how far, on average, scores vary from the mean, and the **variance** is equal to the standard deviation squared.
- Other examples of measures of variability include the **range** (the difference between the highest and the lowest scores) and the **interquartile** range (the difference between the scores at the 25th and 75th percentiles of a distribution).

The sample data set

Participant	Gender	Mathexam	College	Satquant
1	1	24	1	570
2	1	18	2	450
3	2	34	1	600
4	2	27	2	450
5	1	15	1	580
6	1	26	1	550
7	1	42	2	480
8	2	25	1	520
9	2	31	1	450
10	2	44	2	550

The Data

- For each of the participants in Figure above, their **gender**, score on a math entrance exam (**mathexam**), whether they were attending college at home or away (**college**), and their score on the math portion of the SAT (**satquant**) were recorded.
- **Gender** and **college** are categorical variables, and **mathexam** and **satquant** are continuous variables, with higher scores indicating better performance on both measures (the mathexam scores range from 0 to 50 and satquant scores range from 200 to 800).
- For **gender**, males are assigned a “1” and females are assigned a “2”; for **college**, those attending college at home are assigned a “1” and those attending college away are assigned a “2.”

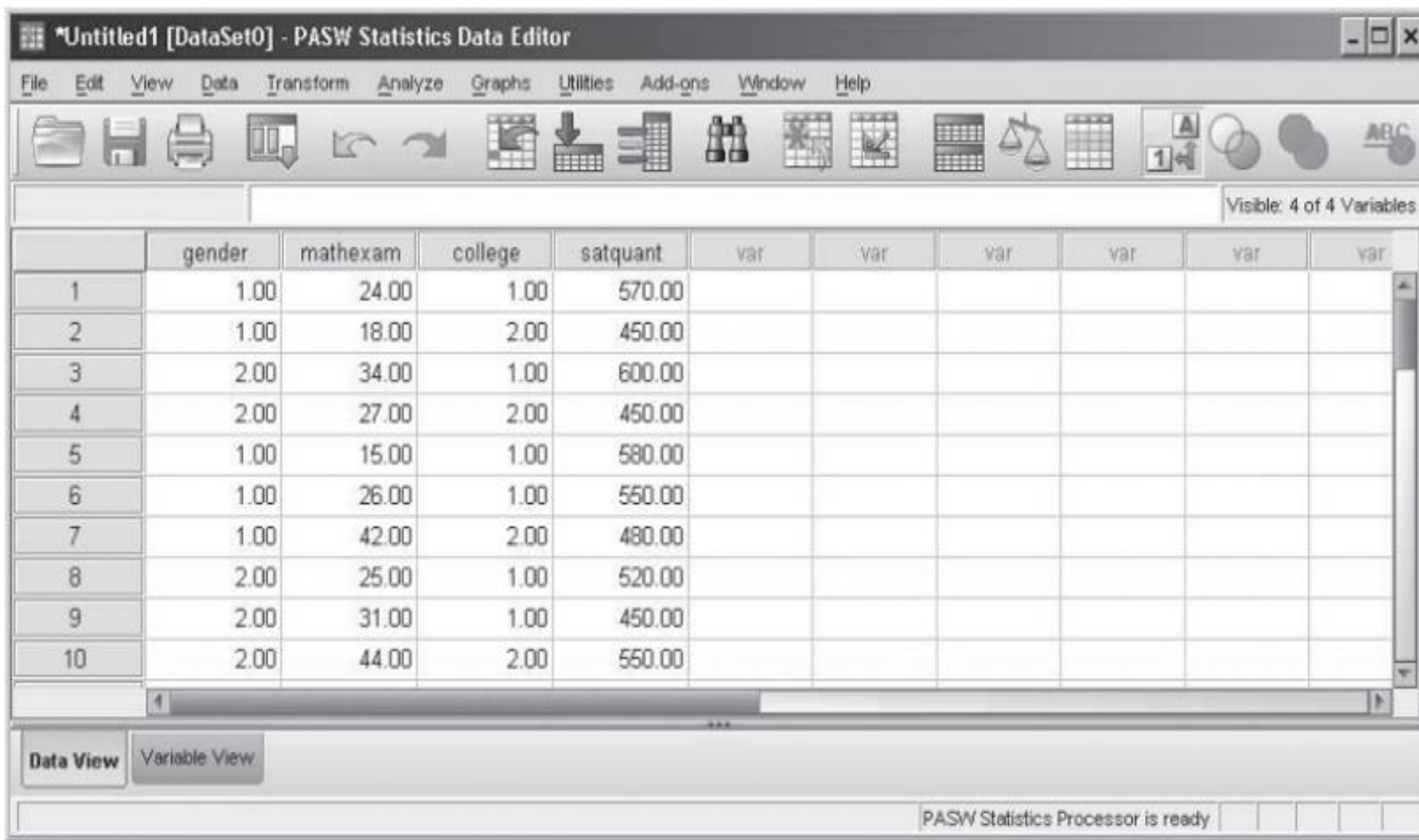
Create the Variables

1. Start SPSS.
2. Click the Variable View tab. In SPSS, four variables will be created. The variables will be named gender, mathexam, college, and satquant, respectively.
3. Using the process described previously, enter the variable names gender, mathexam, college, and satquant, respectively, in the first four rows of the Variable View window

Prior to entering the data, we'll create value labels in SPSS for the categorical variables gender and college.

4. Using the process described previously, create value labels for gender and college. For gender, 1="male" and 2="female" For college 1= "home" , and 2="away"

Enter the Data



The screenshot displays the PASW Statistics Data Editor interface. The title bar reads '*Untitled1 [DataSet0] - PASW Statistics Data Editor'. The menu bar includes File, Edit, View, Data, Transform, Analyze, Graphs, Utilities, Add-ons, Window, and Help. The toolbar contains various icons for file operations, data manipulation, and analysis. The main data grid shows 10 rows of data with 4 visible variables: gender, mathexam, college, and satquant. The status bar at the bottom indicates 'PASW Statistics Processor is ready'.

	gender	mathexam	college	satquant	var	var	var	var	var	var
1	1.00	24.00	1.00	570.00						
2	1.00	18.00	2.00	450.00						
3	2.00	34.00	1.00	600.00						
4	2.00	27.00	2.00	450.00						
5	1.00	15.00	1.00	580.00						
6	1.00	26.00	1.00	550.00						
7	1.00	42.00	2.00	480.00						
8	2.00	25.00	1.00	520.00						
9	2.00	31.00	1.00	450.00						
10	2.00	44.00	2.00	550.00						

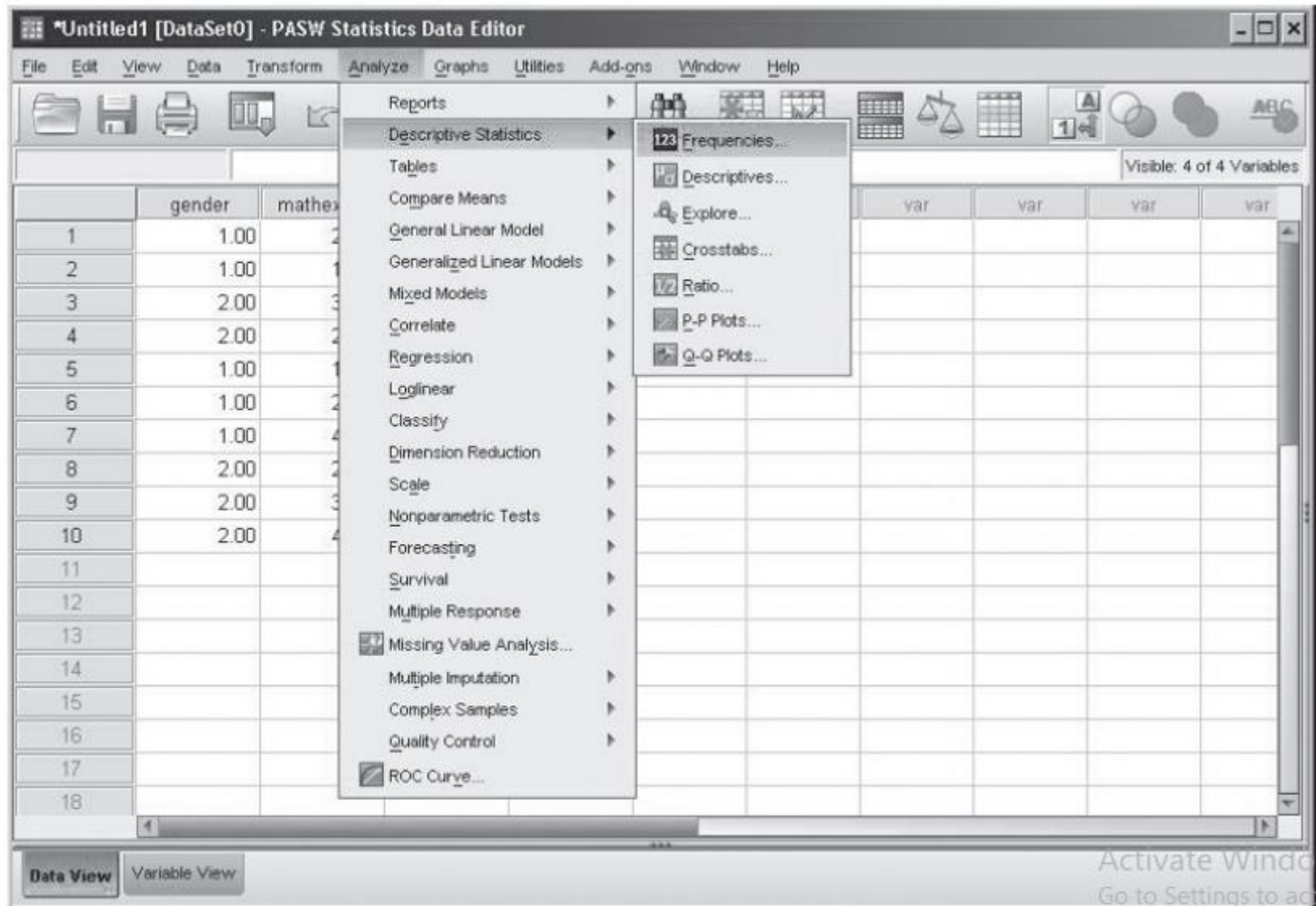
Analyze the Data

- In analyzing the data, first we'll calculate frequencies for the categorical variables **gender** and **college**, and then we'll calculate measures of central tendency and variability for the continuous variables **mathexam** and **satquant**.

Analyze the Data - Frequencies

- To calculate frequencies for gender and college, from the menu bar, select
 1. **Analyze>Descriptive Statistics>Frequencies**

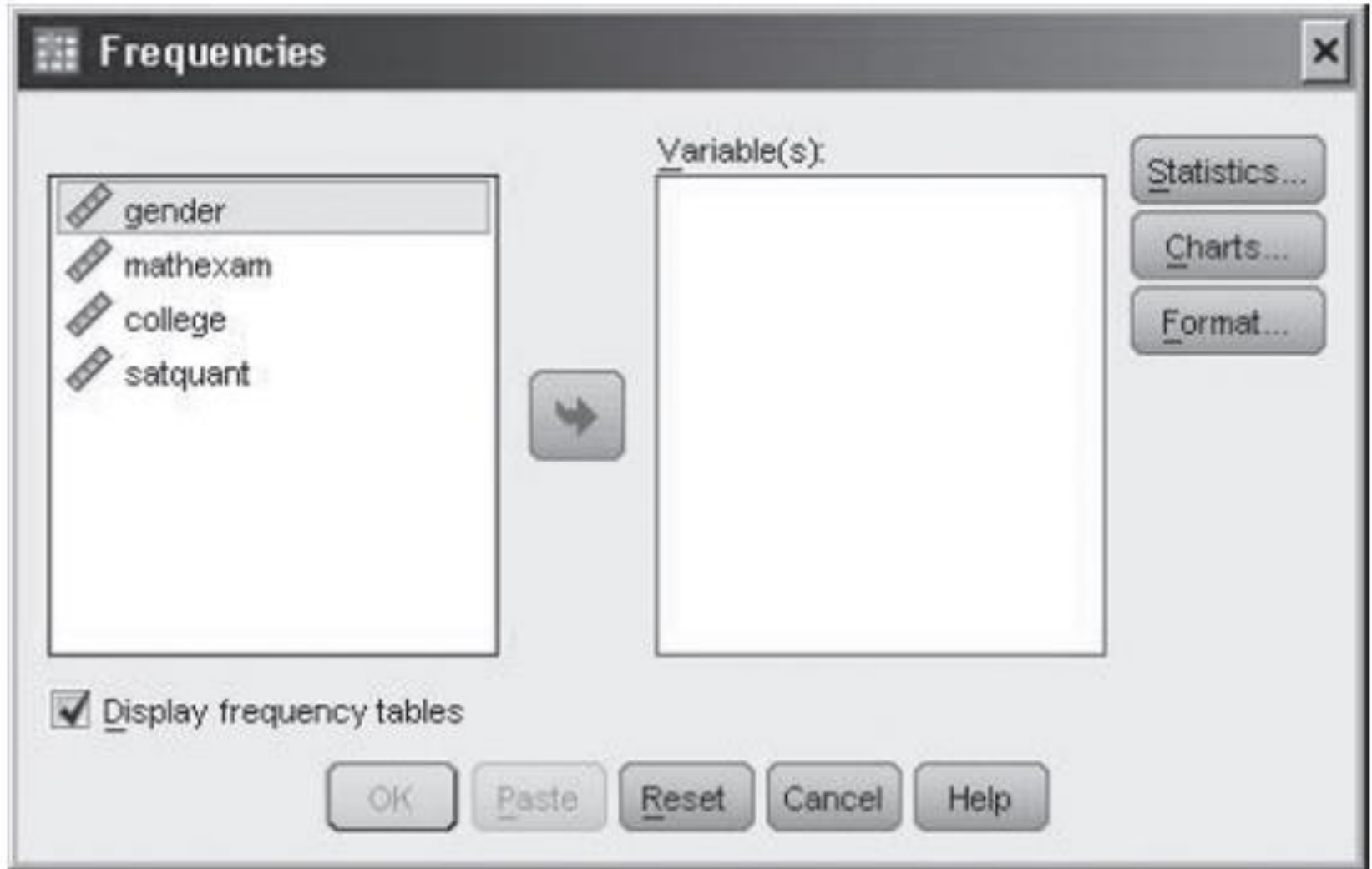
Menu commands for the Frequencies procedure



Analyze the Data cont'd

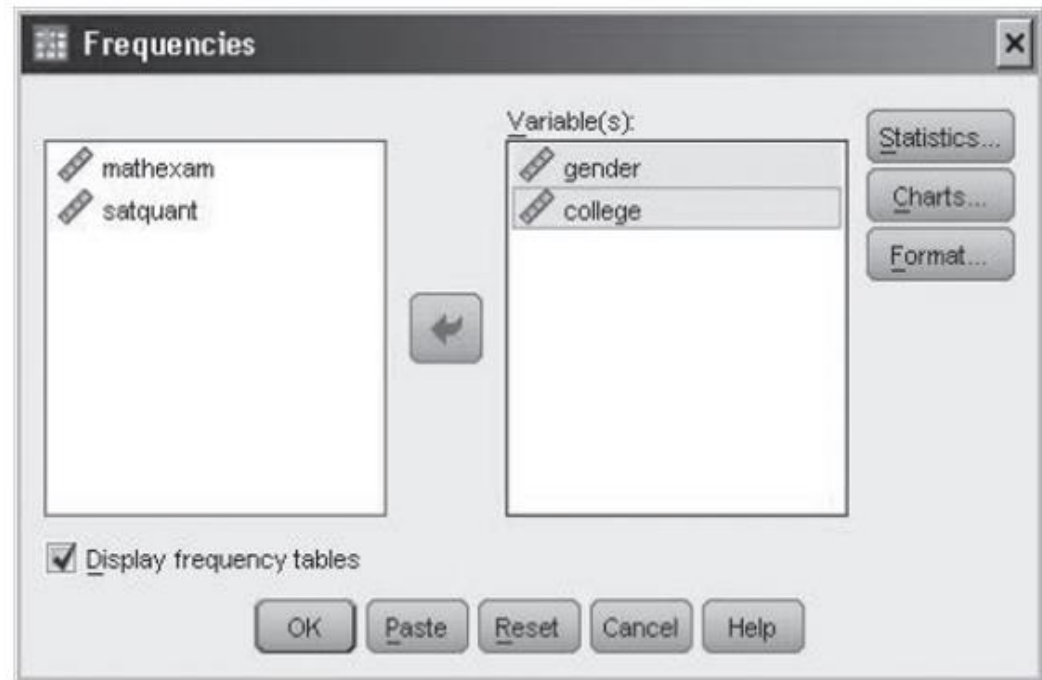
- A Frequencies dialog box opens with the variables **gender**, **mathexam**, **college**, and **satquant** on the left-hand side of the dialog box

The Frequencies dialog box



The Frequencies dialog box

- Select the variable gender and then press and hold down the Ctrl key (the Ctrl key is located in the bottom left-hand corner of your keyboard).
- Select the variable college (gender and college should now be selected). Click the right-arrow button () to move the two variables to the Variable(s) box
- Click OK



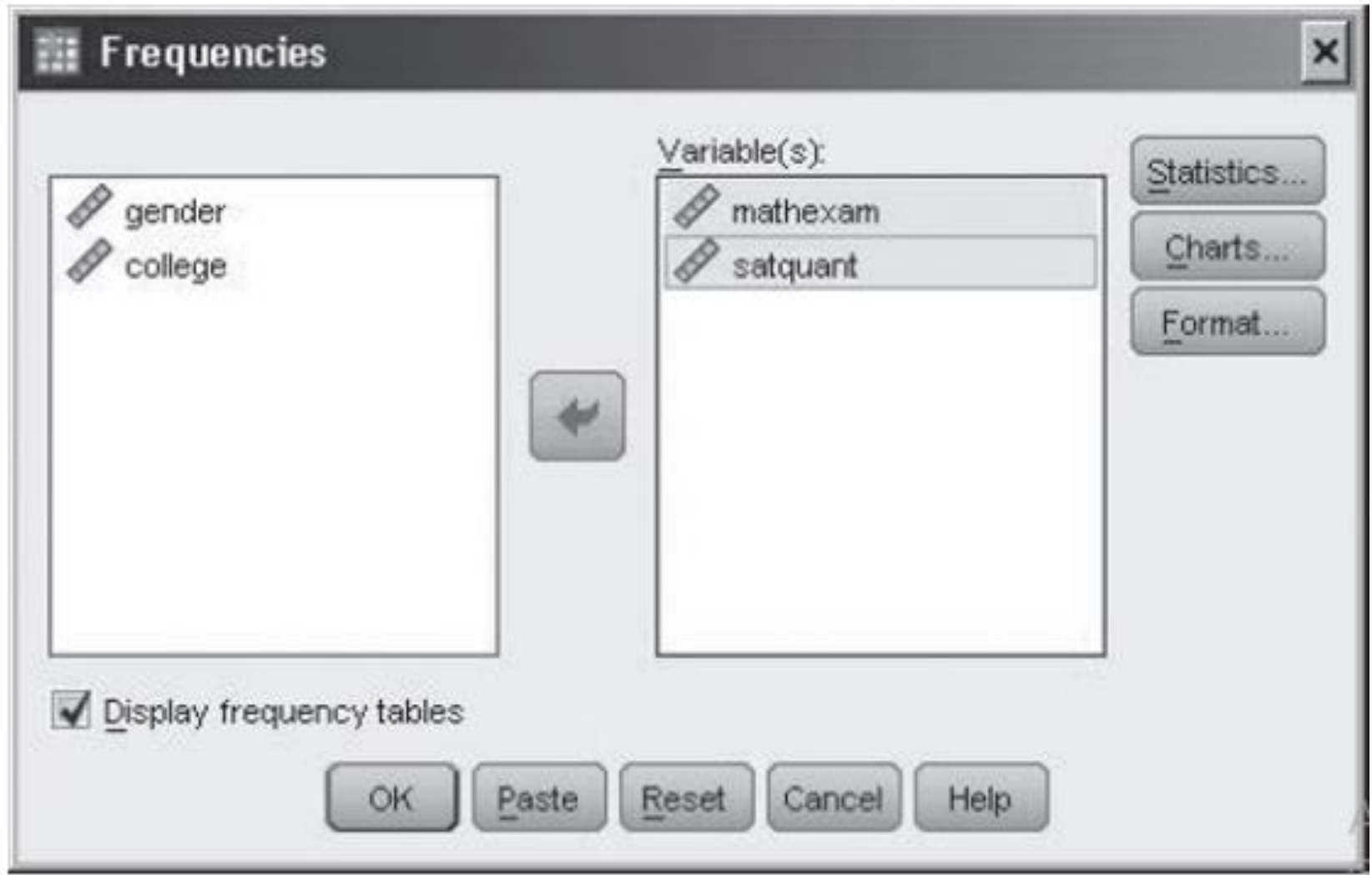
Frequencies cont'd

- The Frequencies procedure runs and the results are presented in the Viewer window.
- Prior to discussing the results of the Frequencies procedure for gender and college, we'll obtain descriptive statistics for mathexam and satquant.

Measures of Central Tendency and Variability

- Next we'll obtain measures of central tendency and variability for the continuous variables **mathexam** and **satquant** using the Frequencies procedure
- To obtain measures of central tendency and variability
 - From the menu bar select **Analyze>Descriptive Statistics>Frequencies**
 - The Frequencies dialog box opens. Click the Reset button to clear the existing variables from the Variable(s) box.
 - Select the variable mathexam and then press and hold down the Ctrl key. Select the variable satquant (both mathexam and satquant should now be selected). Click the right-arrow button () to move the two variables to the Variable(s)

The Frequencies dialog box (continued)



Measures of Central Tendency and Variability cont'd

3. Click the *Statistics* button. The *Frequencies: Statistics* dialog box opens. Under *Central Tendency*, select *Mean*, *Median*, and *Mode*. Under *Dispersion* select *Std. deviation*, *Variance*, *Range*, *Minimum*, and *Maximum*. See Figure 2.8 for details.
 4. Click *Continue*.
 5. In the *Frequencies* dialog box, click on the check mark to the left of *Display frequency tables* to turn this option off. The box should now be empty, indicating that frequency tables will *not* be produced for the variables **mathexam** and **satquant** (frequency tables are typically not produced for continuous variables, as measures of central tendency and variability are more informative). See Figure 2.9 for details.
 6. Click *OK*.
- The Frequencies procedure runs in SPSS and the results are presented in the Viewer window

The Frequencies: Statistics dialog box

Frequencies: Statistics

Percentile Values

- ☐ Quartiles
- ☐ Cut points for: 10 equal groups
- ☐ Percentile(s):

Add
Change
Remove

Central Tendency

- ☒ Mean
- ☒ Median
- ☒ Mode
- ☐ Sum

☐ Values are group midpoints

Dispersion

- ☒ Std. deviation
- ☒ Variance
- ☒ Range
- ☒ Minimum
- ☒ Maximum
- ☐ S.E. mean

Distribution

- ☐ Skewness
- ☐ Kurtosis

Continue Cancel Help

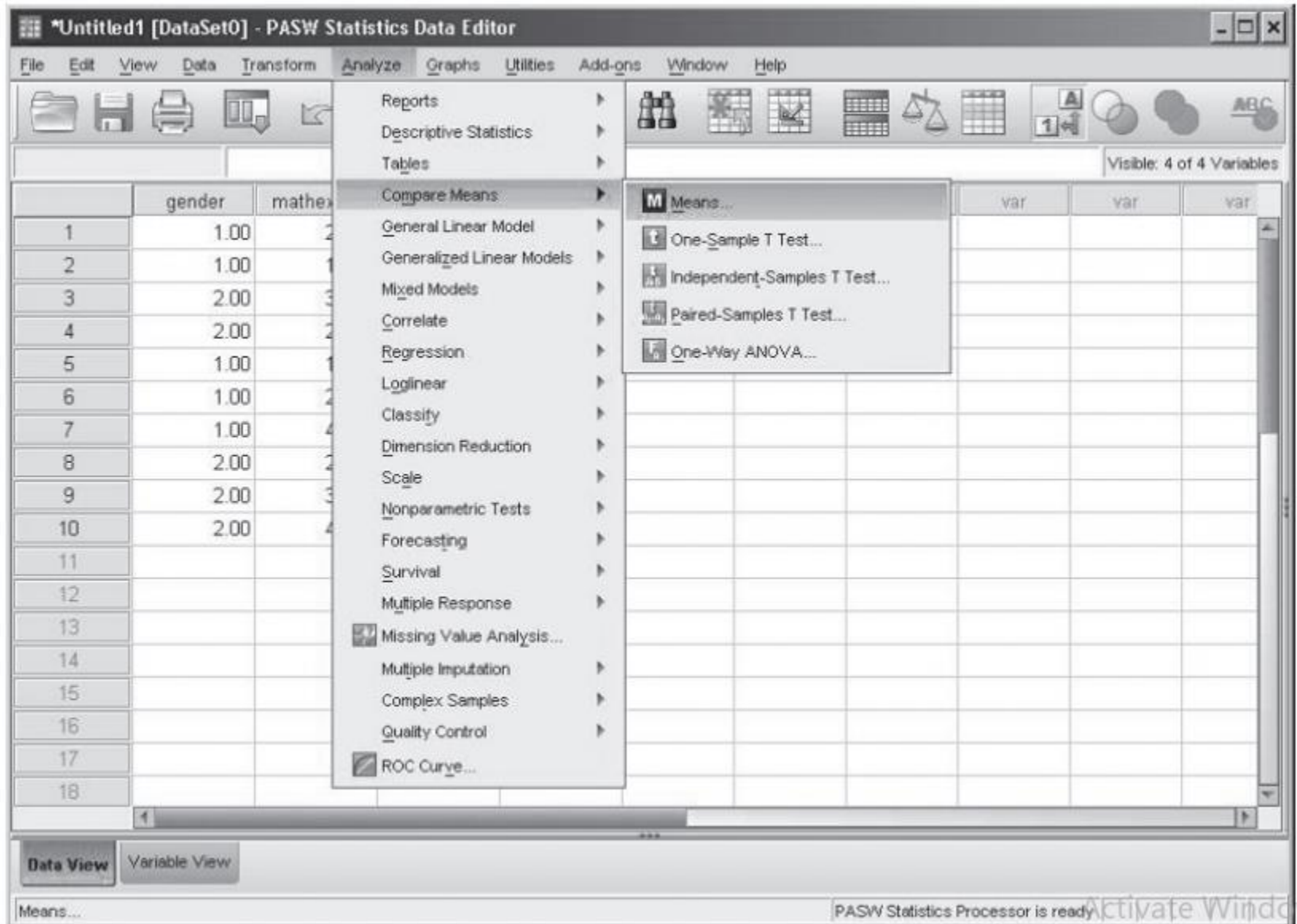
Analysis of Groups Using the Means Procedure

- The Means procedure is useful for obtaining descriptive statistics separately for each group of a variable (or variables) of interest.
- For example, instead of obtaining the mean **satquant** score for all participants, the mean **satquant** scores could be obtained separately for those who attended **college** at home and those who attended **college** away

Analysis of Groups Using the Means Procedure cont'd

- To obtain the means procedure
 1. Make the Data View window active (select **Window>Untitled1 [DataSet0] –IBM SPSS Statistics Data editor**)
 2. From the menu bar, select **Analyze>Compare Means>Means**
 3. The Means dialog box opens
- The variable that the mean and standard deviation will be obtained on (**satquant**) goes into the Dependent List box and the variable that contains the different groups (**college**) goes into the Independent List box

Menu commands for the Means procedure



The Means dialog box



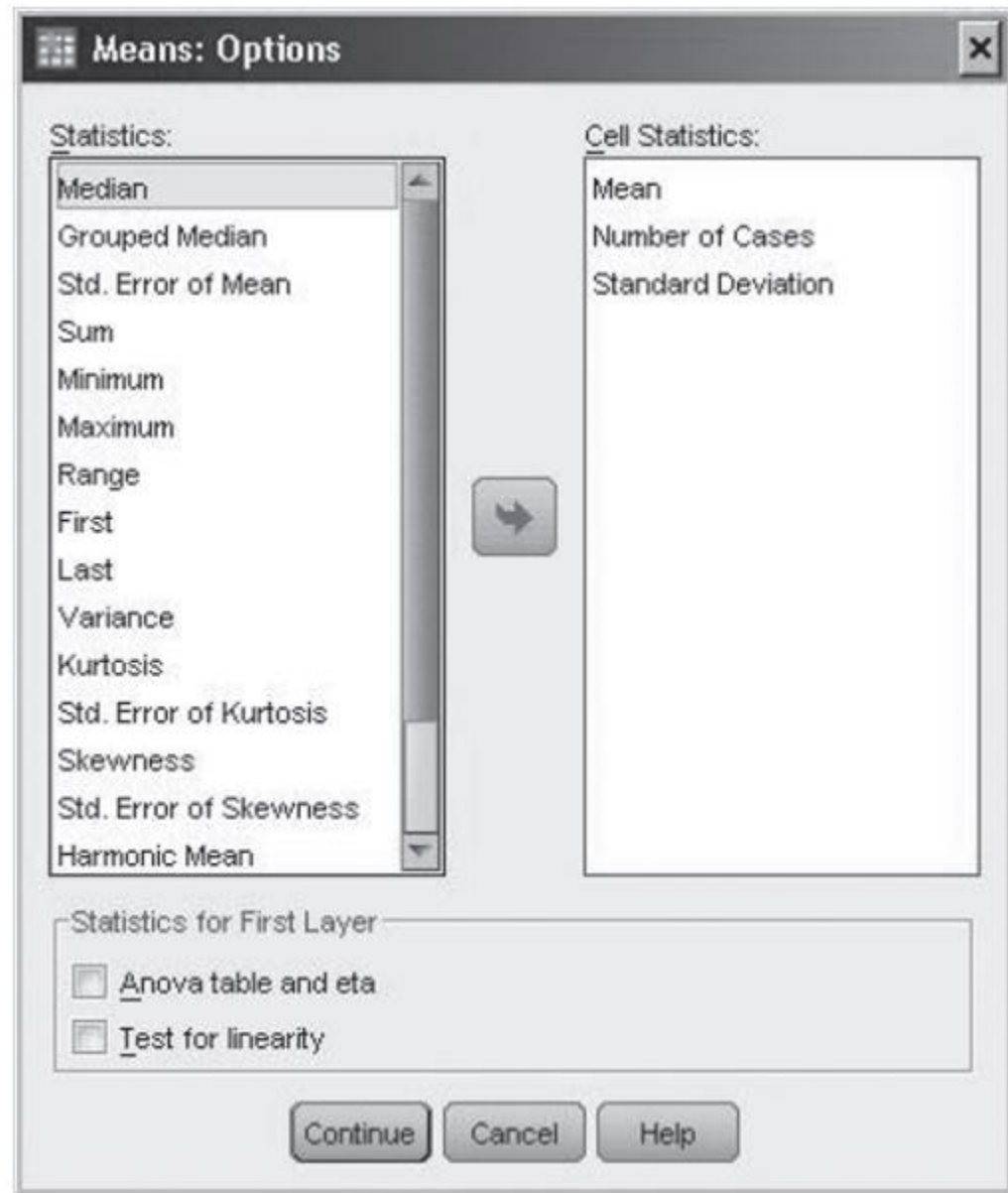
Analysis of Groups Using the Means Procedure cont'd

4. Select satquant and click the upper right-arrow button () to move it to the Dependent List box.
5. Select college and click the lower right-arrow button () to move it to the Independent List box
6. Click Options. The Means: Options dialog box opens

Analysis of Groups Using the Means Procedure cont'd

- On the left-hand side of the dialog box (labeled Statistics) are a number of statistics that can be selected, and on the right-hand side (labeled Cell Statistics) are the statistics that are calculated by default.
 - Since the Mean and Standard Deviation are on the right-hand side already, we'll leave the default options in place. (Number of Cases could be moved to the Statistics box by selecting it and clicking the arrow button, but we'll leave it there since it won't adversely affect our results.)
7. Click Continue.
 8. Click OK.

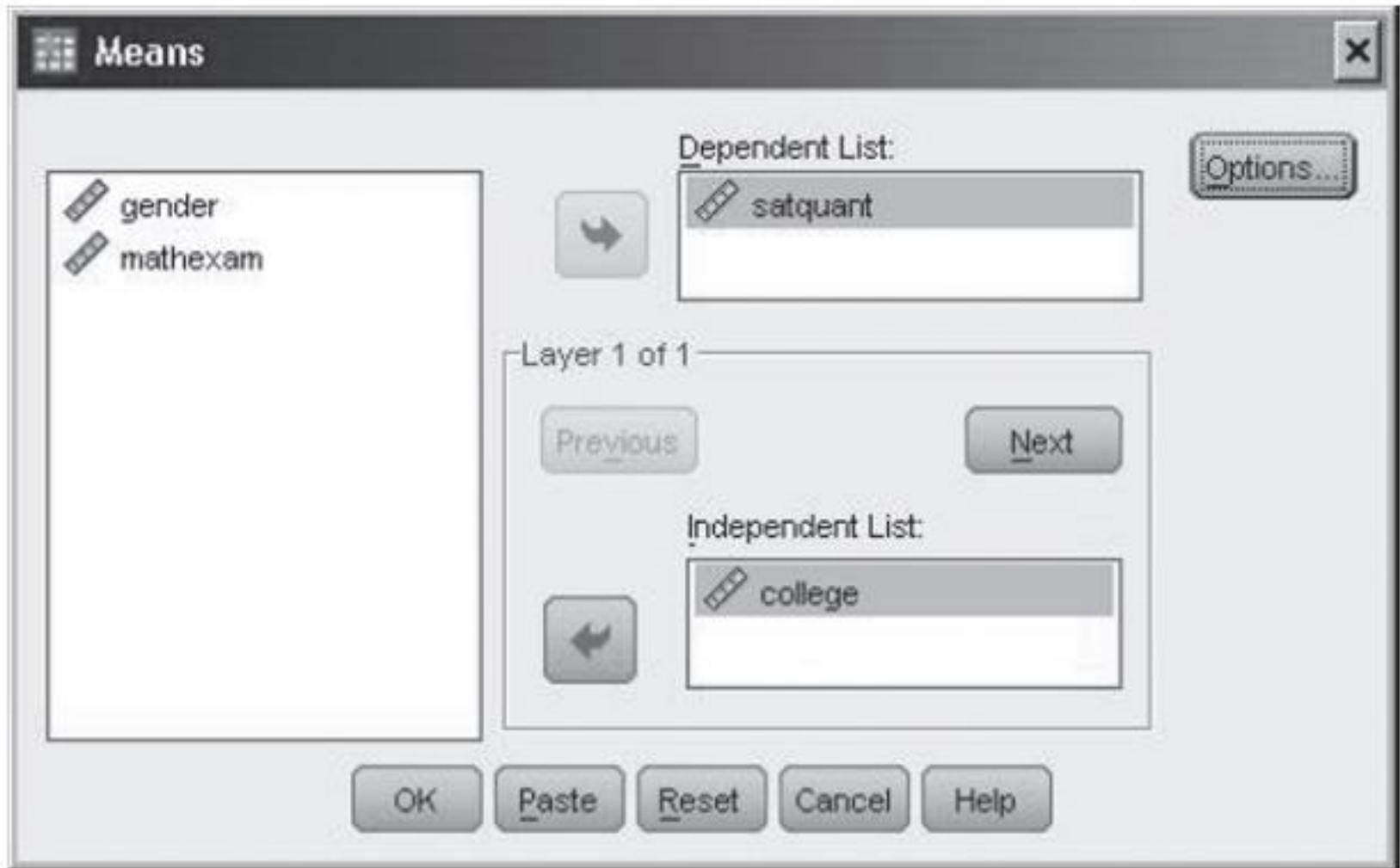
The Means: Options dialog box



Means procedure using the layered option

- To perform the Means procedure using the layered option
 1. Make the Data View window active (select **Window>Untitled1 [DataSet0] –IBM SPSS Statistics Data editor**)
 2. From the menu bar, select **Analyze>Compare Means>Means**
 3. The Means dialog box opens with **satquant** in the Dependent List box and **college** in the Independent List box

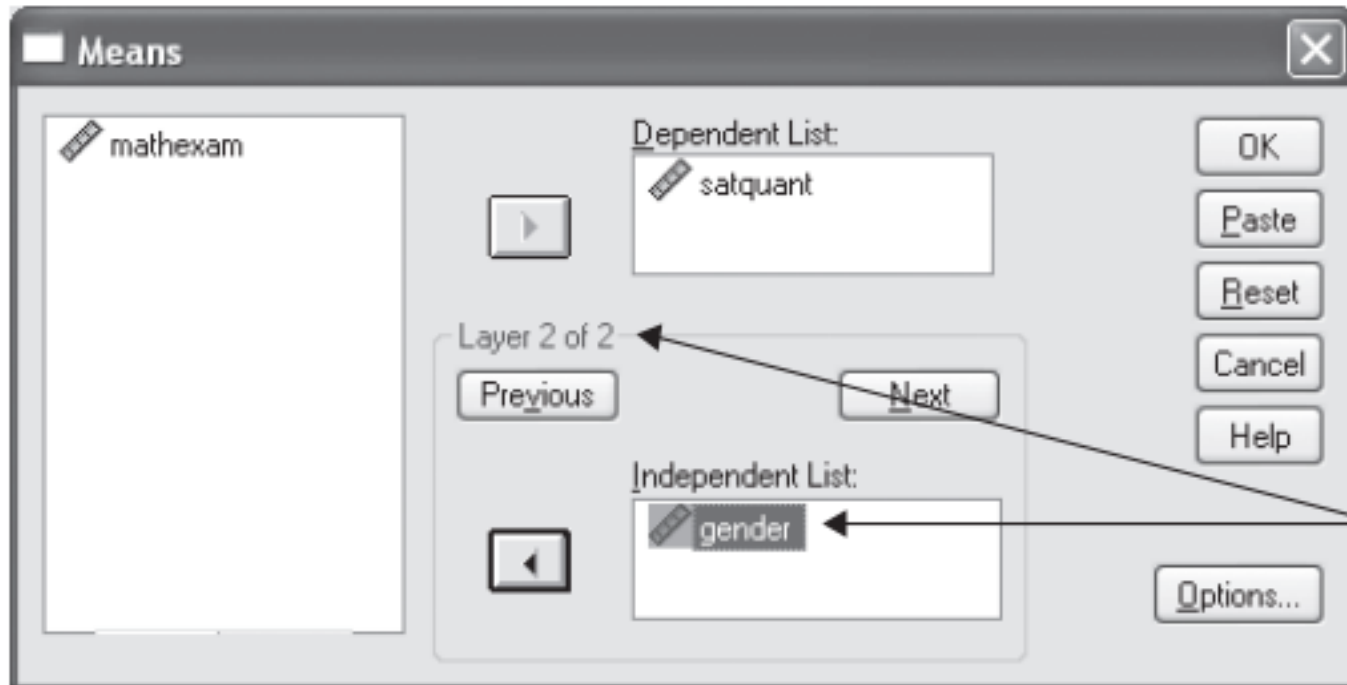
The Means dialog box



Means procedure using the layered option cont'd

4. To layer the results by gender, in the Layer 1 of 1 section of the dialog box, click the Next button (after clicking on Next, you should see 'Layer 2 of 2' above the Previous button). Select gender and click the lower-right arrow button () to move it to the Independent List box
5. Click OK

The Means dialog box using the layered option



Layer 2 of 2 indicates that the results will be layered by the variable entered into the *Independent List* box (**gender**).

Interpret the results

- The Means procedure runs in SPSS and the results are presented in the Viewer window.
- Next we'll discuss the output for the four analyses we conducted. As a review, the four analyses were:
 1. Frequencies on the variables **college** and **gender**.
 2. Measures of central tendency and variability on **mathexam** and **satquant**.
 3. Means and standard deviations on **satquant** for the different **college** groups.
 4. Means and standard deviations on **satquant** for the different **college** groups layered by **gender**

Analysis 1: Frequencies Procedure— Gender and College

Frequencies

Statistics

		gender	college
N	Valid	10	10
	Missing	0	0

Frequency Table

gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	male	5	50.0	50.0	50.0
	female	5	50.0	50.0	100.0
	Total	10	100.0	100.0	

college

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	home	6	60.0	60.0	60.0
	away	4	40.0	40.0	100.0
	Total	10	100.0	100.0	

Frequencies—Statistics

- The Statistics table displays the sample size for the variables **gender** and **college**.
- The table indicates that there were 10 valid observations for **gender** and **college** (all 10 people had values on both variables).
- The value of zero in the Missing row indicates that there were no missing values for either of the variables (a missing value for **gender**, for example, would indicate that the **gender** of one of the participants was not recorded).

Frequency Table—Gender

- The frequency table for **gender** indicates the frequency, or number of participants, for each category of **gender**.
- In the table there are five males and five females, with each **gender** representing 50% of the data set.

Frequency Table—College

- The frequency table for college indicates the frequency for each category of **college**.
- The table shows that six students attended **college** at home and four students attended **college** away from home, representing 60% and 40% of the data set, respectively

Analysis 2: Frequencies Procedure— Mathexam and Satquant

- The Statistics table displays the results of the Frequencies procedure for the continuous variables **mathexam** and **satquant**. (Recall that we requested measures of central tendency and variability for these variables.)
- The mean and standard deviation for **mathexam** is 28.60 and 9.38 (rounding to two decimal places) and for **satquant** is 520.00 and 58.31, respectively.
- The minimum and maximum values indicate the lowest and highest values in the data set and are helpful for ensuring that all the data are within the possible range of values (a value outside of this range would most likely indicate that a data entry error has been made).
- The minimum and maximum values of 15.00 and 44.00 for **mathexam** and 450.00 and 600.00 for **satquant** show that there were no out-of-range values for either variable (as indicated earlier, the range of possible scores for **mathexam** is 0 to 50 and for **satquant** is 200 to 800)

Output of the Frequencies procedure for mathexam and satquant

Frequencies

Statistics

		mathexam	satquant
N	Valid	10	10
	Missing	0	0
Mean		28.6000	520.0000
Median		26.5000	535.0000
Mode		15.00 ^a	450.00
Std. Deviation		9.38320	58.30952
Variance		88.044	3400.000
Range		29.00	150.00
Minimum		15.00	450.00
Maximum		44.00	600.00

Analysis 3: Means Procedure— Satquant by College

Means

Case Processing Summary

	Cases					
	Included		Excluded		Total	
	N	Percent	N	Percent	N	Percent
satquant * college	10	100.0%	0	.0%	10	100.0%

Report

satquant

college	Mean	N	Std. Deviation
home	545.0000	6	53.94442
away	482.5000	4	47.16991
Total	520.0000	10	58.30952

Case Processing Summary

- The Case Processing Summary table indicates the number of observations included in the analysis (10), the number of observations excluded (0), and the total number of observations in the data set (10).
- The table indicates that all 10 participants were included in the analysis (all 10 people had values on **satquant** and **college**)

Report

- The Report table shows the mean, sample size (N), and standard deviation for the college groups home, away, and total (home and away combined).
- Inspecting the means for the two groups shows that the six students who attended **college** at home had a higher mean **satquant** score (545.00) than the four students who attended **college** away (482.50).
- Those who stayed at home for **college** also had **satquant** scores that were slightly more variable than those who attended **college** away (the standard deviation for home was 53.94 compared to a standard deviation of 47.17 for away).
- The mean **satquant** score for the entire sample (Total) was 520.00 with a standard deviation of 58.31 (this agrees with the values obtained in our second analysis).

Analysis 4: Means Procedure—Satquant by College Layered by Gender

Means

Case Processing Summary

	Cases					
	Included		Excluded		Total	
	N	Percent	N	Percent	N	Percent
satquant * college * gender	10	100.0%	0	.0%	10	100.0%

Report

satquant

college	gender	Mean	N	Std. Deviation
home	male	566.6667	3	15.27525
	female	523.3333	3	75.05553
	Total	545.0000	6	53.94442
away	male	465.0000	2	21.21320
	female	500.0000	2	70.71068
	Total	482.5000	4	47.16991
Total	male	526.0000	5	57.70615
	female	514.0000	5	65.03845
	Total	520.0000	10	58.30952

Case Processing Summary

- The Case Processing Summary table indicates the number of observations included in the analysis (10), the number of observations excluded (0), and the total number of observations in the data set (10).
- The table indicates that all 10 participants were included in the analysis (all 10 people had values on **satquant**, **college**, and **gender**).

Report

- The Report table displays the mean, sample size (N), and standard deviation on **satquant** for **college** layered by gender.
- The first section of the table displays the mean and standard deviation for males and females who attend **college** at home.
- Notice that males have a higher mean score than females on **satquant** (566.67 vs. 523.33) and that females have a larger standard deviation (75.06 vs. 15.28).
- The second section of the table displays the mean and standard deviation for males and females who attend **college** away from home.
- For this group of students, females have a higher mean than males on **satquant** (500.00 vs. 465.00) and a larger standard deviation (70.71 vs. 21.21).

Report cont'd

- Finally, the last section of the table displays the mean and standard deviation for males and females for home and away combined.
- When the **college** groups are combined, males have a higher mean **satquant** score than females (526.00 vs. 514.00), while females have a larger standard deviation (65.04 vs. 57.71).

Activity

- A soccer coach evaluated the impact of two different kicking methods (a toe kick and a heel kick) on the effectiveness of kicking a soccer ball. Four boys and four girls kicked a ball using one of the two methods and the distance (in feet) and accuracy (on a 1 to 10 scale) of the kicks were recorded.
- The experience level of each child was also ascertained, with higher values indicating greater experience. Half of the children (two boys and two girls) kicked the ball using a toe kick and the other half kicked the ball using a heel kick. The data are provided in Figure below
- For gender, **1=“boys”** and **2=“girls”** For method, **1=“toe kick”** and **2 =“heel kick”**

Activity cont'd

Gender	Method	Experience	Distance	Accuracy
1	1	5	50	5
1	1	6	45	4
1	2	2	35	6
1	2	9	40	9
2	1	2	42	3
2	1	7	39	6
2	2	4	28	7
2	2	2	32	6

Activity cont'd

- Enter the data in SPSS as shown in figure above and perform the appropriate analyses to answer each of the questions below. Enter value labels as appropriate for **gender** and **method**.
 - a. Report the frequencies for gender and method.
 - b. Report the mean and standard deviation for **experience**, **distance**, and **accuracy** for the entire sample
 - c. Report the mean and standard deviation for **experience**, **distance**, and **accuracy** separately for boys and girls. Which group has a higher mean on each of the three variables? Which group has a larger standard deviation?

Activity cont'd

- d. Report the mean and standard deviation for **experience**, **distance**, and **accuracy** separately for the toe kicking and heel kicking methods. Which group has a higher mean on each of the three variables? Which group has a larger standard deviation?
- e. Report the mean and standard deviation for **experience**, **distance**, and **accuracy** for gender layered by method. Which of the four conditions has the lowest mean experience and accuracy scores? Which has the highest mean kicking distance?



*Live as if you were
to die tomorrow.
Learn as if you were
to live forever.*

Mahatma Gandhi